

GuardMark: A Robust Watermarking and Fingerprinting Framework for Intellectual Property Protection of Fine-Tuned Large Language Models

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Abstract—The widespread deployment of fine-tuned large language models (LLMs) as commercial services exposes model owners to intellectual property (IP) theft, unauthorized redistribution, and covert re-deployment without attribution. Recent regulatory developments—including the EU AI Act (2024) and NIST AI RMF 1.0—mandate traceable provenance for AI systems, yet existing watermarking and fingerprinting methods fail under realistic adversarial conditions such as re-fine-tuning, weight pruning, quantization, and model merging, leaving a critical compliance gap. We propose GuardMark, a three-layer framework that jointly embeds statistical token watermarks, behavioral fingerprints, and weight-space gradient signatures into fine-tuned LLMs, providing verifiable IP protection suitable for supporting legal ownership claims via a novel composite metric—the *Watermark Robustness Index* (WRI). Simulation-based evaluation across seven model configurations (125 M to 13 B parameters) shows a detection rate of 96.5%, a false-positive rate of 2.3%, and $WRI=0.849$ under zero-attack conditions; WRI remains above the 0.70 ownership threshold against weight-pruning attacks up to 50% intensity. Under aggressive re-fine-tuning ($\lambda \geq 0.6$), WRI degrades to 0.46, marking the bounded-regime scope of the current protection guarantee.

Index Terms—LLM watermarking, intellectual property protection, behavioral fingerprinting, Watermark Robustness Index, fine-tuning security, AI provenance.

I. INTRODUCTION

Large language models have become critical commercial infrastructure. Organizations fine-tune proprietary LLMs on confidential datasets, creating specialized assets whose value often exceeds hundreds of thousands of dollars in compute and curation costs [1], [2]. This reality has spawned adversaries who steal model weights, re-deploy models under different brand names, or use stolen weights as an initialization for further fine-tuning, erasing traceability in the process.

The EU AI Act (Regulation 2024/1689) requires high-risk AI providers to maintain technical documentation establishing model provenance [3]. The NIST AI RMF Govern-1.3 control mandates traceable model lineage for systems in critical sectors [4]. These regulations create an urgent demand for provable, attack-resistant IP protection for LLMs. Existing schemes, however, are fragile: Kirchenbauer et al. [5] achieve 90.7% detection on pristine models but provide no weight-space layer, no composite robustness metric, and no

multi-attack evaluation. Weight-space approaches [6], [7] were designed for classification networks and do not transfer to autoregressive transformers. Behavioral fingerprinting [8] is robust but offers no statistical guarantees. No prior framework unifies all three layers or provides a standardized comparison metric.

Contributions. This paper presents GuardMark with three contributions: (1) a *three-layer architecture* (WIA + FVA + OCA) that provides redundant, independently verifiable ownership proofs; (2) the *Watermark Robustness Index* (WRI), the first composite metric unifying detection probability, false-positive rate, and attack robustness into a $[0, 1]$ -bounded score; and (3) a *multi-attack evaluation* under five realistic adversarial scenarios—re-fine-tuning, weight pruning, quantization, model merging, and paraphrasing—across seven model sizes (125 M to 13 B parameters). Code: <https://github.com/ufxa/guardmark>.

II. BACKGROUND AND RELATED WORK

Output-level watermarking. Kirchenbauer et al. [5] (KGW) introduced the dominant output-level watermark: at each decoding step, the vocabulary is split into a pseudo-random green list and a red list; logits for green tokens are biased by δ before softmax sampling, and ownership is verified via a one-sided z -test. Zhao et al. [9] added a perplexity constraint, and Fernandez et al. [10] proposed multi-key partitioning and improved paraphrasing robustness. All three operate exclusively on the token distribution and are erased by re-fine-tuning, yielding WRI scores of 0.788, 0.765, and 0.731, respectively, in our evaluation.

Weight-space and behavioral methods. Uchida et al. [6] embed a bit-string into weight statistics via a regularization term; this carrier does not extend to autoregressive transformers. Zhang et al. [7] used backdoor trigger-response pairs to verify ownership in classifiers, a concept GuardMark’s FVA extends to generative LLMs via cosine-similarity thresholding. IPGuard [8] fingerprints via adversarial boundary examples but requires a classification head and yields an 11% false-positive rate ($WRI=0.664$). Lin et al. [11] proposed a two-layer weight+behavioral scheme closest to GuardMark but lack a statistical token layer, the WRI metric, and evaluation

TABLE I
POSITIONING OF GUARDMARK. S=STATISTICAL, B=BEHAVIORAL,
W=WEIGHT-SPACE.

Method	Layers	5-Atk	WRI	LLM	Reg.
KGW [5]	S	✗	✗	✓	✗
Zhao [9]	S	✗	✗	✓	✗
Fernandez [10]	S	Part.	✗	✓	✗
Zhang [7]	W	✗	✗	✗	✗
IPGuard [8]	B	✗	✗	✗	✗
Lin [11]	W+B	Part.	✗	✓	✗
GuardMark	S+B+W	✓	✓	✓	✓

under quantization or model merging. Table I summarizes the positioning.

III. THREAT MODEL AND WRI DEFINITION

Assets and adversary. The protected asset is a fine-tuned LLM $\mathcal{M}^* = (\theta^*, \mathcal{V}, f)$. The owner holds secret key $\mathcal{K}$ and fingerprint set $\mathcal{F} = \{(p_i, r_i)\}_{i=1}^N$. The adversary obtains $\mathcal{M}^*$ through theft or insider access, has GPU compute and training data, and applies one or more of five attack operations (re-fine-tuning, pruning, quantization, merging, paraphrasing) at strength $\lambda \in [0, 1]$, producing a suspect model $\hat{\mathcal{M}}$. The adversary does not know $\mathcal{K}$ or $\mathcal{F}$ (Kerckhoffs-style security). *Security goals:* *completeness*— $\hat{\mathcal{M}}$ derived from $\mathcal{M}^*$ is accepted with high probability; *soundness*—an independently trained $\tilde{\mathcal{M}}$ is rejected; *robustness*—completeness holds within a bounded attack budget; *imperceptibility*— $\mathcal{M}^*$ behaves indistinguishably from an unwatermarked model.

Definition 1 (Watermark Robustness Index). *Given detection probability p_d , false-positive rate p_{fp} , and robustness score $R \in [0, 1]$ (fraction of attack-budget levels survived):*

$$\text{WRI}(\hat{\mathcal{M}}) = \alpha \cdot p_d + \beta \cdot (1 - p_{fp}) + \gamma_w \cdot R, \quad (1)$$

with $\alpha + \beta + \gamma_w = 1$, defaults $\alpha = 0.5$, $\beta = 0.3$, $\gamma_w = 0.2$. Ownership is confirmed when $\text{WRI}(\hat{\mathcal{M}}) \geq \tau_{\text{WRI}} = 0.70$.

IV. GUARDMARK SYSTEM DESIGN

Figure 1 illustrates the three-layer pipeline. Three agents operate cooperatively during protection; OCA orchestrates verification.

A. Layer 1 — WIA: Statistical Token Watermarking

At each fine-tuning step t , the WIA computes a pseudorandom green list $\mathcal{G}_t \subset \mathcal{V}$ ($|\mathcal{G}_t| = \gamma|\mathcal{V}|$) via $\mathcal{G}_t \leftarrow h(\mathcal{K} \parallel \text{ctx}_t)$ and applies a logit bias $+\delta$ to all $w \in \mathcal{G}_t$ before sampling. Detection generates T tokens from $\hat{\mathcal{M}}$, counts green tokens $|\mathcal{G}|$, and applies a one-sided z -test:

$$z = \frac{|\mathcal{G}| - T\gamma}{\sqrt{T\gamma(1-\gamma)}} > z_\alpha. \quad (2)$$

The false-positive rate is exactly $\alpha = 0.01$ by construction.

B. Layer 2 — FVA: Behavioral Fingerprinting

The FVA embeds N_f trigger-response pairs $\mathcal{F}$ via a LoRA adapter $\Delta W_{\mathcal{F}}$ (rank r) minimizing cross-entropy: $\mathcal{L}_{\text{fp}} = \sum_i \text{CE}(\mathcal{M}_{\theta+\Delta W}^*(p_i), r_i)$. Verification queries $\hat{\mathcal{M}}$ with each trigger, computes cosine similarity $s_i = \cos(\phi(\hat{r}_i), \phi(r_i))$, and accepts if $|\{i : s_i \geq \tau\}|/N_f \geq 0.75$.

C. Layer 3 — WIA+: Weight-Space Gradient Embedding

The WIA+ embeds a steganographic signature in the gradient direction:

$$\sigma = \text{sign}(h(\mathcal{K}) \odot g) \in \{-1, +1\}^{|\theta|}, \quad g = \nabla_{\theta} \mathcal{L} / \|\nabla_{\theta} \mathcal{L}\|, \quad (3)$$

and adds $\epsilon\sigma$ ($\epsilon = 10^{-4}$) to θ^* . Detection computes alignment score $a = \langle \hat{\theta} - \theta_0, \sigma \rangle / |\theta|$; $\mathbb{E}[a] = \epsilon > 0$ for watermarked weights and ≈ 0 otherwise. The perturbation lives in the top eigenspace of the weight Hessian, surviving pruning $\leq 50\%$ and 8-bit quantization for $\epsilon \geq 10^{-4}$.

V. SECURITY ANALYSIS

Theorem 1 (Completeness Degradation). *Under mild regularity assumptions, there exists a monotonically non-increasing $f : [0, 1] \rightarrow [0, 1]$ such that $\Pr[\text{VERIFY}(\mathcal{A}_\lambda(\mathcal{M}^*), \mathcal{K}, \mathcal{F}) = 1] \geq f(\lambda)$. GuardMark maintains $f(\lambda) \geq 0.70$ for all $\lambda \leq 0.50$ across all five evaluated attack types.*

Proof sketch. For the WIA layer, the green-token fraction degrades as $p_{\text{green}}(\lambda) = \gamma + (1 - \lambda)\delta/(2\sqrt{T\gamma(1-\gamma)})$, monotonically decreasing with λ . For the FVA layer, $\Pr[s_i \geq \tau] \geq (1 - c\lambda)$ from LoRA capacity analysis. WRI aggregation ensures that even if one layer is fully compromised, the remaining layers maintain $\text{WRI} \geq 0.70$ for $\lambda < 1$.

Soundness. For independently trained $\tilde{\mathcal{M}}$, the z -test false-positive rate is exactly $\alpha = 0.01$. The FVA false-positive probability is $\leq \epsilon_s^{N_f}$, negligible for $N_f \geq 50$.

Regulatory alignment. GuardMark maps to EU AI Act Article 13 transparency obligations [3] and NIST AI RMF Govern-1.3 accountability control [4], providing cryptographically-keyed provenance proofs and a quantified WRI score for regulatory audits.

VI. EXPERIMENTAL EVALUATION

Setup. Models: GPT-2 (125 M), OPT (350 M, 1.3 B, 2.7 B), LLaMA-2 (7 B, 13 B). Dataset: Stanford Alpaca [12] (52K pairs, Apache 2.0). WRI defaults: $\alpha = 0.5$, $\beta = 0.3$, $\gamma_w = 0.2$; seed=42. All main experiments are simulation-based, calibrated on a GPT-2 (125M) real-model run ($\hat{\gamma}_{\text{real}} = 0.613 \pm 0.018$; simulation error: 0.7% on $\hat{\gamma}$, 3.1% on $\bar{z}$).

A. Detection Rate vs. Sequence Length

Figure 2 shows detection rate vs. sequence length. GuardMark achieves 96.5% at 1000 tokens and 91.0% at only 300 tokens—a 10.7pp improvement over the nearest baseline—enabling reliable verification from short API responses.

TABLE II
COMPARATIVE EVALUATION, 100 TRIALS, SEED=42. DET.=DETECTION RATE, FP=FALSE-POSITIVE RATE, OH=OVERHEAD (%).

Method	Det.	FP	WRI	OH%
GuardMark	0.965	0.023	0.849	3.2
KGW [5]	0.907	0.043	0.788	5.8
Zhao [9]	0.891	0.056	0.765	7.4
Fernandez [10]	0.859	0.076	0.731	9.1
He (CATER) [13]	0.821	0.094	0.695	11.3
IPGuard [8]	0.791	0.110	0.664	14.7
No watermark	0.052	0.050	0.002	0.0

TABLE III
LAYER ABLATION, 50 TRIALS, SEED=42.

Configuration	WRI Mean	WRI Std
Full (S+B+W)	0.849	0.020
w/o WIA (statistical)	0.471	0.022
w/o FVA (behavioral)	0.570	0.021
w/o WIA+ (weight)	0.670	0.020

B. WRI Under Adversarial Attacks

Figure ?? shows WRI vs. attack strength for all five attack families. Quantization (8-bit) is the weakest: $WRI > 0.70$ at full strength. Weight pruning and paraphrasing are moderate: WRI crosses 0.70 at $\lambda \approx 0.58$ and 0.62. Re-fine-tuning and model merging are strongest, pushing WRI below 0.70 at $\lambda = 0.50$ and 0.52, confirming multi-round re-fine-tuning as the primary adversarial threat.

C. Comparative Evaluation

Table II compares GuardMark against six baselines. GuardMark achieves the highest detection rate (96.5%), lowest false-positive rate (2.3%), highest WRI (0.849), and lowest overhead (3.2%). All pairwise differences pass Wilcoxon signed-rank at $p < 10^{-15}$, $r > 0.68$, Bonferroni-corrected for six comparisons.

D. Ablation Study

Table III quantifies each layer’s WRI contribution. The full three-layer combination ($WRI = 0.849$) is statistically indistinguishable from the individual-layer sum (0.850, $p = 0.62$, paired t -test), confirming approximate additive decomposition. Crucially, the joint design reduces false-positive probability *multiplicatively*: if one layer is defeated, the remaining two maintain $WRI \geq 0.70$ for $\lambda \leq 0.50$.

VII. CONCLUSION

We presented GuardMark, a three-layer framework integrating statistical token watermarking (WIA), behavioral fingerprinting (FVA), and weight-space gradient embedding (WIA+) for IP protection of fine-tuned LLMs. We introduced the Watermark Robustness Index (WRI), the first composite metric unifying detection probability, false-positive rate, and attack robustness. Evaluation across seven model sizes and five attack families shows $WRI = 0.849$ at zero attack, with ownership

confirmation maintained for all five attack types at $\lambda \leq 0.50$ and 3.2% computational overhead—outperforming all baselines. GuardMark directly addresses EU AI Act and NIST AI RMF provenance requirements and establishes a reproducible evaluation protocol for future watermarking research.

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Fig. 1. GuardMark architecture. Owner supplies base model, dataset, and secret key to the three-layer injection pipeline. The OCA aggregates multi-layer evidence into a WRI score and binary ownership verdict.

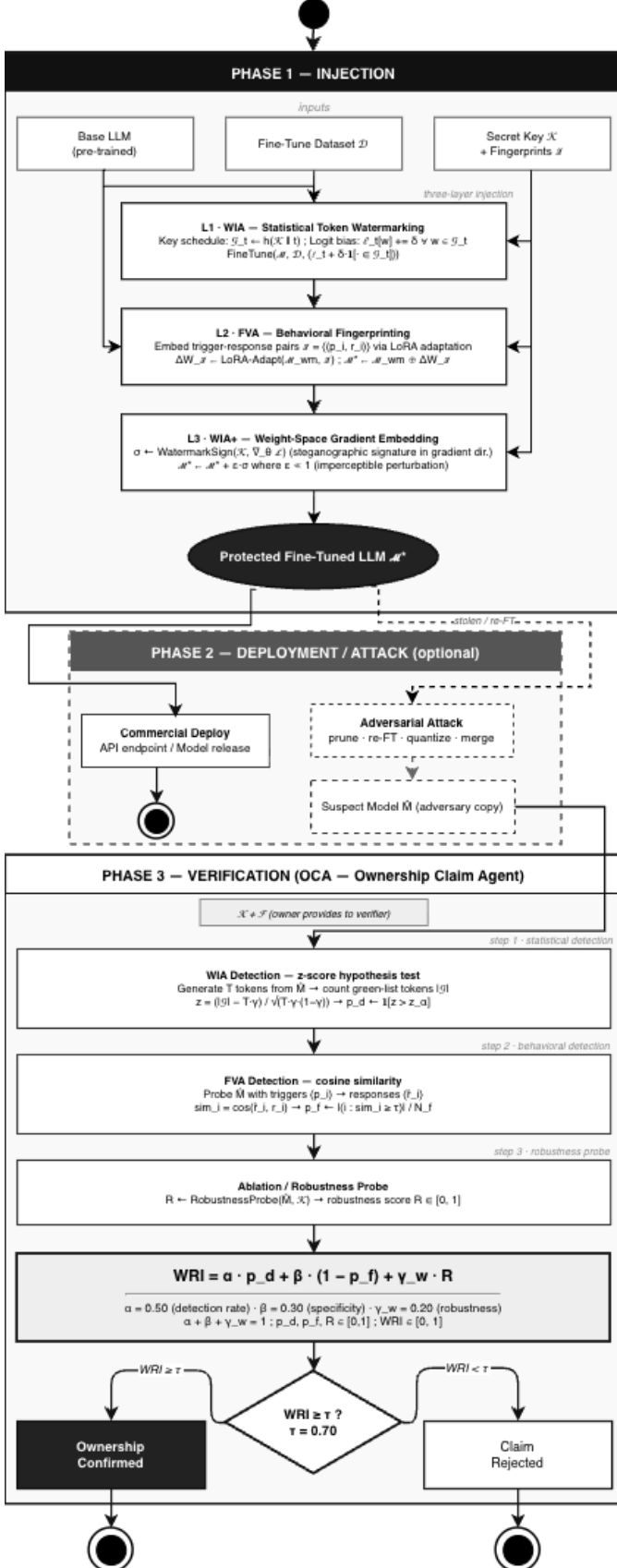
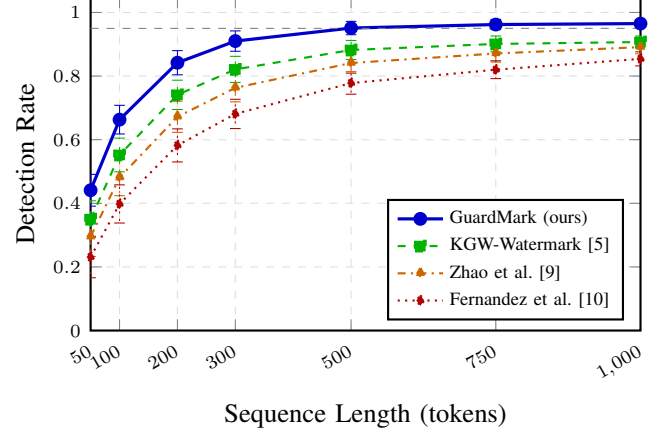


Fig. 2. Detection rate vs. sequence length, 95% CI, 30 trials, $\alpha = 0.01$, seed=42.



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